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PERSEKUTUAN
62530 PUTRAJAYA**

KERTAS PENERANGAN (*INFORMATION SHEET*)

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TAJUK/TITLE: **ENVIRONMENTAL HAZARD AND PRECAUTION**

TUJUAN/PURPOSE:

This Information Sheet is intended to describe on the ability to demonstrate level of understanding in Health, Safety and Environment adaptation and apply it to work activities. The person who is competent in this CA shall be able to adapt to HSE by executing his/her duties while observing Health, Safety and Environment Rules and Regulations.

PENERANGAN/INFORMATION:

1. Condition And Nature Of Environmental At Workplace

Environment involves in matters that affect pollution of atmosphere, global warming and other dangers which effect the planet by nature and human activities. Nature discusses the

earth wide variety of lives and plants and its endangered species. Environmental condition attributed by the following:

- i. Pollution : undesirable state of the natural environment being contaminated with harmful substances as a consequence of human activities.
- ii. Erosion :condition in which the earth's surface is worn away by the action of water and wind.
- iii. Deforestation :the state of being clear of trees.
- iv. Depopulation :the condition of having reduced numbers of inhabitants.
- v. Climate :the weather in some location averaged over some long period of time.
- vi. Glaciations :the condition of being covered with glaciers or masses of ice; the result of glacial action.
- vii. Inhospitableness :the environmental condition in a region that lacks a favourable climate or terrain for life or growth.
- viii. Meteorological Condition : condition - the prevailing environmental conditions as it influences the prediction of weather.

There are many things one can do to conserve the environment. Observe routine work practice in a way towards environmental friendly activities. Enhance knowledge and awareness for self sustainability.

2. Environmental Consciousness And Precaution

Everybody is making strides to protect and preserve the environment. True enough, the new war cry of most individuals and organizations today is "Go Green". Our decisions and actions contribute to the deterioration or well-being of the environment, and it is something that none of us can ever afford to look right through. Now is the right time to make this planet safe both for the present generation and for the generations to come. As stewards of this planet, we should understand that it is our lives that are put at stake when we pollute the environment and over-consume our resources. Each choice we make has an environmental impact, so it is imperative that we learn how to make the right ones. Here are ways to be environmentally conscious:

Recycling is an indispensable part of environmental preservation. Among the items that can be recycled are papers, clothes, bulbs, electronics and plastic. It is best to have separate disposal bins for each of these items so waste segregation won't be a problem. Before disposal, look for those that can be recycled. This saves such items from filling up landfills and emitting harmful chemicals during decomposition.

Instead of trashing your old items, donate them. Aside from recycling, reusing is another good way to cut down energy expenditure that comes with recycling and production. Among the items you can donate are old furniture, shoes, electronics and clothes. That way, you don't only get to protect the environment but also make an underprivileged individual's life easier.

Typically, electricity comes from non-renewable fossil fuels, which release green house gases like carbon dioxide into the earth's atmosphere. When you conserve electricity at home by using energy-efficient appliances, you contribute to the reduction of global warming. You can do this by turning off your air conditioner when it is not too hot or when you unplug the charger of your laptop when its battery is still full.

One of the best ways through which you can use water responsibly at home is by repairing your leaky faucets. After use, make sure that your faucets are closed enough not to leak. You can also run your laundry just when you have a full load so you won't have to use much volume of water every day. In watering your lawn, use a sprinkler instead of a hose and don't water your lawn when it rains.

Instead of buying inorganic compost for your garden, make your own. You can do this by using your kitchen waste such as fruit and vegetable peels; dry leaves, or flowers. By using organic compost, you not only avoid waste from clogging up landfills but you also convert them into effective fertilizers.

Fields typically concerned by the precautionary principle are the possibility of:

- Global warming or abrupt climate change in general
- Extinction of species
- Introduction of new and potentially harmful products into the environment, threatening biodiversity (e.g., genetically modified organisms)

- Threats to public health, due to new diseases and techniques (e.g., AIDS transmitted through blood transfusion)
- Long-term effects of new technologies (e.g. health concerns regarding radiation from cell phones and other electronics communications devices)
- Persistent or acute pollution (asbestos, endocrine disruptors...)
- Food safety (e.g., Creutzfeldt-Jakob disease)
- Other new bio-safety issues (e.g., artificial life, new molecules)

The precautionary principle is often applied to biological fields because changes cannot be easily contained and have the potential of being global. The principle has less relevance to contained fields such as aeronautics, where the few people undergoing risk have given informed consent (e.g., a test pilot). In the case of technological innovation, containment of impact tends to be more difficult if that technology can self-replicate. Bill Joy emphasized the dangers of replicating genetic technology, nanotechnology, and robotic technology in his article in Wired Magazine, "Why the future doesn't need us", though he does not specifically cite the precautionary principle. The application of the principle can be seen in the public policy of requiring pharmaceutical companies to carry out clinical trials to show that new medications are safe.

3. Environmental Hazards

There are several types of environmental pollutions that leading to environmental hazards and risks.

Action has been taken to establish a more environmentally conscious society. Manufacturers turn into alternative source of energy with less pollution discharged. There are also moves toward encouraging individuals to start paying more attentions on self-sustainability.

3.1 Air Pollution

Air is polluted by many factors. Beside automobiles exhaust and large industrial factories, deforestation and various agricultural and industrial practices,, natural factors such as volcanic activity and forest wildfires also attributed to it. These activities have led to increased atmospheric concentrations of a number of greenhouse gases, including carbon dioxide, methane, nitrous oxide, chlorofluorocarbons, and ozone in the lower part

of the atmosphere. Any sort of contaminant introduced into the atmosphere will disturb the chemical composition of the atmosphere. Carbon dioxide is one of the most prominent contaminants.

These contaminated gases will trap heat within the earth atmosphere. Hence, boosting up the temperature of the planet overall. The increment in temperature then affects the earth ecosystems, whether due to melting in the polar ice caps or simple seasonal shifts on land. This type of air pollution also called 'thermal pollution'.

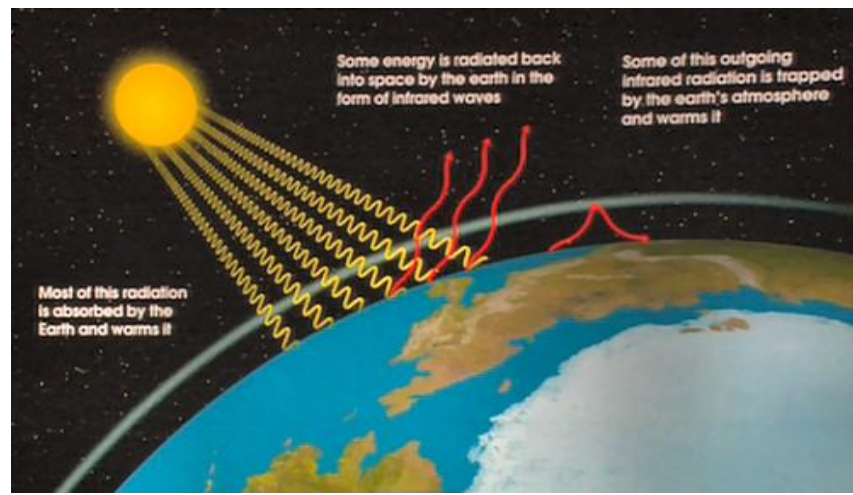


Figure 1: The Green House Effect

Air pollution consequently lead to hazy atmosphere and eventually turn into acid rain which causes cancer, asthma and even death or diminishing of crops due to reduced level of oxygen.



Picture 1: Air Pollution – Industrial Factory

Malaysia has hit by haze phenomenon. The haze we see in our city skyline is caused by tiny particulates suspended in the atmosphere. At high concentrations, these particulates

scatter and absorb sunlight resulting in diminished horizontal visibility thereby giving the atmosphere a characteristic opalescent appearance. Haze is not confined to urban environments; it may also be observed in rural areas.

The particles that cause the haze phenomenon can originate from many sources, some of which are natural and some anthropogenic. Natural sources include the oceans, forests and ground surface. However the majority of the particulates are from human activities which include open burning, land clearing, vehicular use and combustion of fossil fuels in industrial boilers.



Open burning

Vehicle emissions

Combustion of fossil fuels

Picture 2: Other source of air pollution

i) Is Haze Harmful?

Haze originating from natural sources such as sea salt and soil dust is generally harmless to humans. Slight, transient haze visible in most towns and cities has not been proven to cause any serious long-term health effects on the population. However, during a severe haze episode, prolonged exposure to high concentrations of particulates can be harmful to health. Associated with previous haze episodes, there have been reports of increases in incidences of eye and throat irritations and respiratory difficulties among sensitive groups. During a haze episode, the public are strongly urged to follow the advice issued by the local health authorities.

ii) How Can We Reduce The Incidence Of Haze?

Firstly, we should refrain from open burning of waste. Most incidences of local haze can be traced to this activity. Motor vehicles, due to their increasing numbers, are also major sources of haze particulates in urban areas. To reduce haze, vehicles should be well maintained. Choose fuels which are the least polluting. Large scale land clearing which exposes large tracts of barren land, not only results in soil

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erosion and degradation, but also provides a source of particulates to the atmosphere. Promoting efforts such as rapid replanting and staggered land clearing can reduce haze. Remember that every individual has a role to play to preserve the quality of our environment as a legacy for our future generations.

3.2 Water Pollution

Water pollution is defined by any contamination of water resources such as rivers, streams, ponds or lakes. Water pollution can strike anywhere and in any form, be it chemicals, bacterial contaminations, or trashes that end up at the bottom of the ocean or floating on river surface.

Surface waters are degraded by both point and non-point sources of pollution. The major point sources of pollution are sewage treatment plants, agro-based industries, manufacturing industries, silage or grey-water from commercial and residential premises, and pig farms. Non-point source (or diffuse) pollution is largely due to storm runoff after a downpour. Earthworks and land clearance activities contribute to siltation of rivers and can be both point and non-point sources of pollution.

Water pollution can also be caused by natural factors, such as dead animals decomposing in water or growing levels of sediment leading to shallower and less substantial bodies of water.

Water pollution is more serious than air contamination as it may threaten the future of the planet. Pollutants in water may reduce the capacity of drinking water and eventually compel a dire impact of the earth population and crops. Hence, source of food is devastated.

3.3 Soil Pollution

Also known as land pollution and is most prominent case. Contaminants in the ground permeate into water sources, thus bringing about higher level of water pollution. However, contamination of ground soils due to oil spills and other hazardous waste dumps or heavy use of dangerous pesticides can seep into every faction of environmental health. Dead animals, crops, and trees will render soil pollution that transform into environmental pollution. Soil pollution can also arise from battery acid and

other chemicals leaking from computers, appliances, and other types of human waste that have not been properly disposed of or recycled.

3.4 Nuclear Pollution

Also known as radioactive pollution and is less said in form of environmental pollution despite being the highest level of danger category.

Nuclear pollution is described by high levels of radiation concentrated in any one place. It could be due to nuclear bomb blasts, nuclear power plant accidents or leakages, improper disposal of radioactive wastes, or etc. As proven in Japan nuclear bombings during the World War II, radiation pollution is incredibly detrimental to both human and animal populations, leading to cancerous effects, birth defects and other horrific health problems.

3.5 Light Pollution, Noise Pollution, and Visual Pollution

These types of pollution are normally the least discussed when it comes to environmental preservation, even though they are important elements of environmental pollution.

i) Light Pollution

It occurs mostly in big cities. Though, it also have an impact in suburban areas, around airports or big sporting event facilities, along highways where illuminated billboards and streetlamps are common.

For humans, light pollution is more of a irritation than an environmental concern, perhaps making it difficult to see the stars or interfering with sleep.

For animals, light pollution can be more prominent, disrupting biological clocks, influencing natural predation patterns, and overall threatening animal habitats located nearby to heavily illuminated signs.

ii) Noise Pollution

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For humans, noise pollution is more of a concern than light pollution. It may lead to disturbance of regular life patterns such as disorder of sleep, distract from work, and causing hearing loss.

It may cause sizable disturbances to animal habitats. It is often tied to light pollution for how it arises from similar sources, including highways, airports, and loud sporting events or concerts, from construction projects, railroads, and etc.

iii) Visual Pollution

Visual pollution can be anything that makes an area less aesthetically pleasing such as billboards and power lines, abandoned buildings, litters along the road side, landfills and etc. In many cases, visual pollution can be just more than an annoyance. However, it can also lead to other environmental pollution problems.

4 How To Combat Environmental Pollution

After knowing the various types of environmental pollution, now it is time to play your roles in combating them in order to make the earth a cleaner and better place to live in.

To prevent air pollution, turn into finding cars with better fuel economy or go for hybrid or electric vehicles reduce such pollution.

To starve off both soil and water pollutions by practicing proper recycle program, especially on electronic devices and hardly decomposed materials. Control level of waste and litter may reduce problems of visual pollution.

There are numerous other ways to fight against the problems of environmental pollution. One of them is to invest in solar energy.

One important thing to remember is that there is only one Earth, so taking care of it is the most important thing we will ever do. Whether you adopt a greener lifestyle for yourself or to preserve the planet for future generations, there could be no more noble pursuit.

5 Environmental Control Act And Regulations

In industrial countries, voluntary environmental agreements often provide a platform for companies to be recognized for moving beyond the minimum regulatory standards and thus support the development of best environmental practice.

In developing countries, such agreements are more commonly used for remedy significant levels of non-compliance with legislations and regulations. The key advantage of its use in developing countries is to build environmental management capacity.

Malaysia adopts both the prevention and pollution control approach in order to protect the environment, as well as to ensure the environment is clean, safe, healthy, and productive. Environmental Quality Act, 1974 is the legislation that is related to the prevention, abatement, control of pollution and enhancement of the environment in Malaysia. The Act restricts the discharge of wastes into the environment in contravention of the acceptable conditions. It was brought into effect in 1974 with the implementation of a subsequent set of specific regulations and laws. To date 38 sets of Regulations and Orders (19 Regulations, 16 Orders and 3 Rule) have been introduced and enforced.

The legislative approach in environmental management and pollution control using the 1974 Environmental Quality Act has been successful in reducing pollution to a certain extent. It has involved pollution control, prevention of pollution and continuous assessment (monitoring) of the environment. There are still many challenges that need to be addressed to achieve a holistic environmental management. Much of the past and present efforts are very much directed against controlling pollution from point sources while non-point pollution probably continued unabated.

6 Environmental Emergency Response And Evacuation Procedures

Many activities in a modern society are connected to the production and use of man-made chemicals. After World War Two intensive work in chemical laboratories resulted in a dramatic increase in the synthesis and isolation of new chemical compounds. In 1942, a mere 600 000 chemical compounds were known. In 1947, this number had increased to 4 million and since then the total number of identified chemical compounds has risen to around 11 million. At many stages during the production, transportation, storage and use of many of these chemicals there is a risk that accidents will occur, both the possibility of occurrence and the severity of possible accidents depending on the chemicals concerned. The largest overall environmental impact of chemicals is due to diffuse and continuous discharges from human activities such as industry, transport and agriculture.

These activities are also the most intensively investigated. Nevertheless, chemicals released through accidents and other sudden, isolated events also affect the environment (and human health). Many severe chemical accidents in recent history have affected the environment (and humans).

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The complexity and increasing flow of chemicals has resulted in more rigorous demands being placed on both authorities and industrial facilities that handle chemicals to estimate the risks involved. Industry is responsible for taking appropriate measures to handle chemicals safely and the authorities are responsible for giving appropriate guidelines.

Environmental Impact Assessment (EIA) is a well-established method, which can be used as a tool to harmonise the granting procedures for planned projects and certain human activities. The purpose of the rules on EIA is to assess possible impacts of projects on the environment as early as possible in the decision-making processes and to ensure a proper information and consultation of the public about the project. This procedure is carried out for certain types of projects that are likely to have significant environmental impacts.

Preparation of environmental management plan is required for formulation, implementation and monitoring of environmental protection measures during and after commissioning of projects. The plans should indicate the details as to how various measures have been or are proposed to be taken including cost components as may be required.

Environmental Management Plan (EMP) translates the EIA approval conditions into action. The EMP document states in explicit terms what actions will be taken, what measures will be instituted, what structures will be built, what will be installed, when the actions will be executed; etc. in order for the project activities to be compliant with the EIA approval conditions. The EMP is a concrete plan of action which is explicit, illustrative, action-oriented, time-bound and definitive. Even though the EMP exhibits all of the above characteristics, the EMP is by nature, a living document which needs to be revised and updated when there exists certain circumstances which demand changes to be made. These factors may include changes to project details and surrounding areas and inadequacy of the control measures to comply with regulatory standards.

An EMP also a site-specific plan developed to ensure that all necessary measures are identified and implemented in order to protect the environment and comply with environmental legislation. It provides the answers to the following important questions:

- What are the likely environmental issues for your site?
- What likely harm these issues can cause to the surrounding environment?
- How will you manage these issues to minimize harm to the environment?

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SOALAN/QUESTION :

1. What we can do to conserve the environment?

2. List down 5 types environmental hazards

3. How to combat environmental pollution?

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